

A certified coordinate-gauge obstruction for causal-order reconstruction in a $2 + 1$ diamond

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<https://github.com/Quantlyra/quantyra-null-cone-lean>

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Abstract

Finite chronological-order samples are invariant under a smooth change of coordinates that transports both chronology and sampling measure. We give an explicit Lean-certified example showing why quotienting density coefficients only by spatial rotations and reflections does not remove this freedom. On the unit $2 + 1$ Minkowski diamond, the standard conformal diamond flow transports normalized flat volume to a nonconstant smooth density between $1/2$ and $3/2$ with Euclidean Lipschitz constant at most two. Every finite labeled and unlabeled directed-order law agrees with the flat law, while the coordinate supremum distance remains positive after every spatial orthogonal transformation. Nevertheless, the volume-normalized Lorentzian metrics are smoothly isometric, with an explicit pullback identity. We prove each step, identify the flow with its established conformal-field-theory formula, and give a statement-to-source map for the accepted formalization. The contribution is an explicit certified instance and an audit of the proposed coordinate gauge. The flow and the general invariance principle are established; no new physical nonidentifiability or priority claim is made.

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1 Question, result and scope

An inverse problem based on finite causal orders must specify which coordinate descriptions represent the same object. In the reconstruction paper [5], a restricted $1 + 1$ density class fixes uniform null-coordinate marginals. A tentative higher-dimensional extension would instead compare density functions on a fixed diamond modulo spatial rotations and reflections. The present example shows that this smaller symmetry group leaves an observable ambiguity inside a smooth, uniformly bounded class.

The obstruction uses an established conformal flow. Its mathematical value here is the complete passage from a Cartesian formula to an admissible density, actual probability laws, a positive coordinate discrepancy and an explicit isometry, together with machine-checked evidence. Section 6 compares the construction with primary sources. We do not treat certification as a proof of originality.

1.1 The fixed model

Write $p = (t, z) = (t, x, y)$, $r = |z|_2$, and set

$$\mathcal{D} = \{(t, z) : |t| + |z|_2 < 1\}, \quad \eta = -dt^2 + dx^2 + dy^2.$$

An overline denotes closure. The underlying coordinate norm is the Euclidean norm on $\mathbb{R}^3$. The chronological relation is the genuine Lorentzian relation

$$p \ll q \iff q_t - p_t > |q_z - p_z|_2. \quad (1)$$

The diamond is convex. Thus the straight segment realizes every relation in (1); conversely, integrating the future timelike cone inequality along a timelike curve gives the same endpoint condition. Consequently this is also the intrinsic chronology of the diamond.

Its flat volume and normalized reference measure are

$$V = \int_{-1}^1 \pi(1 - |t|)^2 dt = \frac{2\pi}{3}, \quad \mu_0 = V^{-1} dt dx dy|_{\mathcal{D}}. \quad (2)$$

Let $\mathcal{K}_3$ consist of real functions smooth on an open neighborhood of $\overline{\mathcal{D}}$ such that

$$\frac{1}{2} \leq \rho \leq \frac{3}{2}, \quad |\rho(p) - \rho(q)| \leq 2\|p - q\|_2 \quad (p, q \in \overline{\mathcal{D}}), \quad \int \rho d\mu_0 = 1. \quad (3)$$

For $\rho \in \mathcal{K}_3$, put $\mu_\rho = \rho\mu_0$ and

$$g_\rho = V^{-2/3} \rho^{2/3} \eta. \quad (4)$$

The Lorentzian volume density of this three-dimensional metric is $V^{-1}\rho$: multiplying η by a positive coefficient c multiplies $\sqrt{|\det \eta|}$ by $c^{3/2}$. Hence $\text{vol}_{g_\rho} = \mu_\rho$ and its total volume is one. Positive conformal scaling preserves time orientation and chronology.

For $n \geq 0$, sample $P_1, \dots, P_n$ independently from μ_ρ and form the matrix

$$A_n(P)_{ij} = \mathbf{1}_{\{P_i \ll P_j\}}.$$

Write ν_n^ρ for its law. This is a Borel pushforward because the inequality in (1) is open. Let $\bar{\nu}_n^\rho$ be the pushforward under simultaneous relabeling of rows and columns. This forgets vertex names and retains the direction of the order; it does not identify an order with its time reverse.

For $R \in \text{O}(2)$, let $T_R(t, z) = (t, Rz)$. The proposed discrepancy is

$$d_{\text{O}(2)}(\rho, \sigma) = \inf_{R \in \text{O}(2)} \sup_{p \in \mathcal{D}} |\rho(p) - \sigma(T_R p)|. \quad (5)$$

The group includes reflections. It does not include nonlinear conformal maps of the diamond.

1.2 Main statement

For $|a| < 1$, define

$$D_a(t, z) = (1 + at)^2 - a^2|z|_2^2, \quad (6)$$

$$N_a(t, z) = (t + a)(1 + at) - a|z|_2^2,$$

$$\Omega_a(t, z) = \frac{1 - a^2}{D_a(t, z)},$$

$$F_a(t, z) = \left(\frac{N_a(t, z)}{D_a(t, z)}, \frac{(1 - a^2)z}{D_a(t, z)} \right). \quad (7)$$

The formulas are used on the positive-denominator neighborhood of the closed diamond. Fix from now on

$$a_* = \frac{1}{100}, \quad \rho_0 = 1, \quad \rho_* = \Omega_{-a_*}^3. \quad (8)$$

Theorem 1.1. *The densities ρ_0, ρ_* belong to $\mathcal{K}_3$. For every $n \geq 0$,*

$$\nu_n^{\rho_*} = \nu_n^{\rho_0}, \quad \bar{\nu}_n^{\rho_*} = \bar{\nu}_n^{\rho_0}.$$

Nevertheless $d_{O(2)}(\rho_0, \rho_) > 0$. The map F_{-a_*} is a smooth time-oriented diffeomorphism of $\mathcal{D}$ satisfying*

$$g_{\rho_*} = F_{-a_*}^* g_{\rho_0}. \quad (9)$$

Both maps and their inverse formulas extend smoothly to neighborhoods of the closed diamond, including the spatial axis and boundary points.

Sections 2–5 prove the theorem. The isometry in (9) identifies the two geometries exactly. The positive discrepancy is a failure of the proposed coordinate gauge to identify two descriptions of that geometry.

2 The conformal diamond flow

Lemma 2.1. *For $|a| < 1$, F_a maps the open and closed diamond bijectively to themselves, with inverse F_{-a} . It is smooth on an open neighborhood of $\bar{\mathcal{D}}$.*

Proof. Use radial null variables $u = t + r$ and $v = t - r$. Membership in $\bar{\mathcal{D}}$ is equivalent to $|u|, |v| \leq 1$, and

$$D_a = (1 + au)(1 + av) \geq (1 - |a|)^2 > 0.$$

Thus $U_a = \{D_a > 0\}$ is an open neighborhood of $\bar{\mathcal{D}}$ and the Cartesian rational formulas are smooth there. There is no division by r in (7).

The function $f_a(s) = (s + a)/(1 + as)$ satisfies

$$f'_a(s) = \frac{1 - a^2}{(1 + as)^2} > 0, \quad f_a(\pm 1) = \pm 1, \quad f_{-a}(f_a(s)) = s.$$

Directly from (7), positivity of Ω_a and $r \geq 0$,

$$t' + r' = f_a(u), \quad t' - r' = f_a(v), \quad r' = \Omega_a r. \quad (10)$$

This proves preservation of the open and closed diamond. When $r > 0$, the spatial direction is unchanged, so applying f_{-a} to both null variables restores t, r, z . At $r = 0$ the spatial coordinates stay zero and the same inverse identity restores t . Applying the argument with $-a$ proves both inverse compositions and surjectivity. The inverse also has a positive-denominator neighborhood, proving the asserted smoothness up to the boundary in the ambient-coordinate sense. $\square$

Lemma 2.2. *For $p \in \bar{\mathcal{D}}$ and $|a| < 1$,*

$$(\mathrm{d}F_a)_p^\top \eta (\mathrm{d}F_a)_p = \Omega_a(p)^2 \eta, \quad \det(\mathrm{d}F_a)_p = \Omega_a(p)^3 > 0. \quad (11)$$

For $p, q \in \bar{\mathcal{D}}$, the separation identity is

$$\eta(F_a(q) - F_a(p), F_a(q) - F_a(p)) = \Omega_a(p)\Omega_a(q)\eta(q - p, q - p). \quad (12)$$

Proof. Away from the axis use polar coordinates, so that $\eta = -du dv + r^2 d\theta^2$ (with the symmetric interpretation of the mixed term). The two null derivatives have product $f'_a(u)f'_a(v) = \Omega_a^2$, and $r'^2 = \Omega_a^2 r^2$ while the angle stays fixed. This proves the bilinear identity there. Cartesian smoothness and continuity extend it to the axis and boundary.

Taking determinants yields $(\det dF_a)^2 = \Omega_a^6$. For a fixed point of $\overline{\mathcal{D}}$, replace a by sa , $0 \leq s \leq 1$. The determinant is continuous and never zero, and is one at $s = 0$. Its sign is therefore positive throughout, proving the determinant formula. This sign step is needed for the density transport below.

For completeness, (12) is an identity of rational Cartesian functions. Multiply by $D_a(p)^2 D_a(q)^2$, insert the three numerators in (7), and expand: the resulting polynomial is

$$(1 - a^2)^2 D_a(p) D_a(q) (-(q_t - p_t)^2 + (q_x - p_x)^2 + (q_y - p_y)^2).$$

Both denominators are positive, giving (12). Appendix A gives a Cartesian derivative certificate which also avoids polar coordinates altogether. $\square$

Proposition 2.3. *For $|a| < 1$ and $p, q \in \overline{\mathcal{D}}$, $p \ll q$ if and only if $F_a(p) \ll F_a(q)$.*

Proof. Suppose $p \ll q$. Along the continuous deformation $s \mapsto F_{sa}$, equation (12) keeps the squared Lorentz separation strictly negative. The transformed time difference cannot be zero, because a vector with zero time component has nonnegative Lorentz square. It starts positive at $s = 0$, hence remains positive at $s = 1$ by the intermediate value theorem. Negative square and positive time difference together give (1) for the images. Apply the same argument to F_{-a} and use Lemma 2.1 for the converse. This proves orientation preservation for arbitrary pairs, including pairs with different spatial directions. $\square$

3 Density bounds and actual measure transport

Lemma 3.1. *The function ρ_* is smooth on a neighborhood of $\overline{\mathcal{D}}$ and satisfies the bounds and Euclidean Lipschitz condition in (3).*

Proof. Smoothness follows from $D_{-a_*} > 0$. Factoring in null variables gives

$$(99/100)^2 \leq D_{-a_*}(p) \leq (101/100)^2, \quad (99/101)^3 \leq \rho_*(p) \leq (101/99)^3.$$

The latter interval lies inside $[1/2, 3/2]$, as follows by multiplying the positive integer denominators.

Here is a direct global Lipschitz estimate using Euclidean distances. Every coordinate of a point of $\overline{\mathcal{D}}$ has absolute value at most one. Hence, for any such p, q and coordinate i ,

$$|p_i - q_i| \leq \|p - q\|_2, \quad |p_i^2 - q_i^2| \leq 2\|p - q\|_2.$$

Expansion of $D_{-a_*} = 1 - 2a_*t + a_*^2(t^2 - x^2 - y^2)$ therefore gives

$$|D_{-a_*}(p) - D_{-a_*}(q)| \leq (2a_* + 6a_*^2)\|p - q\|_2 \leq \frac{3}{100}\|p - q\|_2. \quad (13)$$

If $9/10 \leq d, e \leq 11/10$ and $0 \leq c \leq 1$, then

$$\begin{aligned} \left| \frac{c}{d^3} - \frac{c}{e^3} \right| &= \frac{c|d - e|(d^2 + de + e^2)}{d^3 e^3} \\ &\leq \frac{363/100}{(9/10)^6} |d - e| \leq 7|d - e|. \end{aligned}$$

Take $d = D_{-a_*}(p)$, $e = D_{-a_*}(q)$ and $c = (1 - a_*^2)^3$. Their stated bounds satisfy these hypotheses. Combining this with (13) gives a Lipschitz constant at most $21/100 < 2$. The formal class theorem records the required constant two. $\square$

Proposition 3.2. *For every $|a| < 1$,*

$$(F_a)_*\mu_0 = \Omega_{-a}^3\mu_0. \quad (14)$$

In particular ρ_ is normalized and belongs to $\mathcal{K}_3$, as does 1.*

Proof. The change-of-variables formula for the diffeomorphism $F_a : \mathcal{D} \rightarrow \mathcal{D}$ and Lemma 2.2 give, for every nonnegative Borel function h ,

$$\begin{aligned} \int_{\mathcal{D}} h(F_a(p)) d\mu_0(p) &= \frac{1}{V} \int_{\mathcal{D}} h(q) |\det dF_{-a}(q)| dq \\ &= \int_{\mathcal{D}} h(q) \Omega_{-a}(q)^3 d\mu_0(q). \end{aligned}$$

This proves equality of the actual measures. Setting $h = 1$ proves normalization, rather than assuming it as an additional hypothesis. Lemma 3.1 supplies the remaining conditions for $a = a_*$. The constant function one has all class properties immediately from (2). $\square$

4 Finite order laws

Proposition 4.1. *For every $n \geq 0$, both finite-order laws in Theorem 1.1 agree.*

Proof. Let $P_1, \dots, P_n$ be independent with law μ_0 and define $Q_i = F_{a_*}(P_i)$. Deterministic coordinatewise transport preserves independence, and Proposition 3.2 makes each Q_i have law μ_{ρ_*} . Equivalently,

$$(F_{a_*}^{\times n})_*(\mu_0^{\otimes n}) = \mu_{\rho_*}^{\otimes n}.$$

Proposition 2.3 gives $A_n(Q) = A_n(P)$ pointwise for every sample in $\mathcal{D}^n$. Taking laws proves the labeled equality. Pushing both equal measures through the same relabeling quotient gives the unlabeled equality. At $n = 0$ both laws are the unit mass on the empty order. No estimation, asymptotic limit or time-reversal quotient is used. $\square$

5 Coordinate separation and geometric isometry

Proposition 5.1. *The discrepancy in (5) is positive. More explicitly,*

$$d_{O(2)}(1, \rho_*) \geq \left(\frac{39996}{39601} \right)^3 - 1 > 0. \quad (15)$$

Proof. The density depends on z only through $|z|_2^2$, so $\rho_* \circ T_R = \rho_*$ for every $R \in O(2)$. Each supremum in (5) is thus the same number. It is finite because $|\rho_* - 1| \leq 1/2$. At the interior point $p_* = (1/2, 0, 0)$,

$$\rho_*(p_*) = \left(\frac{1 - 1/10000}{(1 - 1/200)^2} \right)^3 = \left(\frac{39996}{39601} \right)^3 > 1.$$

The supremum bounds this value minus one for every R . Taking the infimum proves (15). This uses an interior point and hence does not depend on whether boundary values enter the coordinate norm. $\square$

Proposition 5.2. *The map $H = F_{-a_*}$ is a smooth time-oriented isometry from $(\mathcal{D}, g_{\rho_*})$ onto $(\mathcal{D}, g_1)$.*

Proof. The previous results give its smooth inverse and time orientation. Since $\Omega_{-a_*} > 0$,

$$\rho_*^{2/3} = (\Omega_{-a_*}^3)^{2/3} = \Omega_{-a_*}^2.$$

Using the derivative pullback identity in (11),

$$H^* g_1 = V^{-2/3} H^* \eta = V^{-2/3} \Omega_{-a_*}^2 \eta = g_{\rho_*}.$$

This equality holds for every tangent-vector pair at every point, including the ambient extension to $\overline{\mathcal{D}}$. It proves the claimed isometry directly. The sampling measures transform in the compatible direction $H_* \mu_{\rho_*} = \mu_0$. $\square$

These propositions complete Theorem 1.1. In particular the nonconstant coordinate coefficient in g_{ρ_*} still describes a flat geometry. A coordinate formula alone does not distinguish Lorentzian isometry classes.

Define, for either the labeled or unlabeled directed laws,

$$\Delta_N(\rho, \sigma) = \max_{1 \leq n \leq N} \text{TV}(\nu_n^\rho, \nu_n^\sigma).$$

Corollary 5.3. *There is no function $b : \mathbb{N} \times [0, 1] \rightarrow [0, \infty)$ with $b(N, 0) \rightarrow 0$ such that, for every pair in $\mathcal{K}_3$ and all $N \geq 1$,*

$$d_{\text{O}(2)}(\rho, \sigma) \leq b(N, \Delta_N(\rho, \sigma)).$$

The assertion holds for either choice of finite-order laws.

Proof. Insert $\rho = 1$ and $\sigma = \rho_*$. Proposition 4.1 gives $\Delta_N = 0$ for every N , while Proposition 5.1 gives a fixed positive lower bound for the left side. Letting N tend to infinity is a contradiction. $\square$

This corollary is an elementary prose consequence of the certified example. It asserts failure of this particular coordinate discrepancy; it supplies no inverse rate for a corrected gauge.

6 Comparison with established work

6.1 The flow is established

Casini, Huerta and Myers give the causal-diamond flow in equation (2.13) of their arXiv v2 PDF [1]; their discussion cites Hislop and Longo [2] for earlier double-cone modular geometry. To identify their formula with ours, set the ball radius to one and $a = \tanh(\pi s)$. Their $x^+ = r + t$ becomes u , and their $x^- = r - t$ becomes $-v$. For $k = e^{-2\pi s}$, elementary algebra gives

$$\frac{(1+w) - k(1-w)}{(1+w) + k(1-w)} = \frac{w+a}{1+aw}, \quad a = \frac{1-k}{1+k}.$$

The opposite sign in the x^- exponential gives the same f_a after replacing x^- by $-v$. Equations (10) then give exactly (7). Thus this manuscript does not introduce a new conformal transformation. Its parameter a_* selects a small member of that established family.

6.2 Probability-law equivalence

Janson’s Theorem 7.1 [3] relates equivalent poset kernels, their finite random-poset laws and measure-preserving representations. Our deterministic kernel is $W(p, q) = \mathbf{1}_{p \ll q}$. Propositions 3.2 and 2.3 supply an explicit measure-preserving order isomorphism between the two measured spaces. The coupling proof is the direct invariance direction of this general framework; it does not provide a new abstract equivalence theorem.

6.3 Reconstruction up to isometry

Braun’s Theorem 1.4 [4] reconstructs finite-volume, causally continuous, future-chronocomplete spacetimes up to smooth isometry from all finite labeled chronological adjacency laws. The weighted version and Remark 1.6 explain the corresponding conformal rescaling. Our explicit pullback identity gives the isometry directly. Positive coordinate distance is therefore compatible with that reconstruction target. We use no unverified application of its converse or its geometric hypotheses. The equality of unlabeled laws here follows from the explicit labeled coupling and makes no claim to solve a general reconstruction conjecture for unlabeled data.

6.4 Contribution and limits of the comparison

The flow, order-law invariance under transport, and geometric interpretation are established ingredients. This paper packages their explicit small-parameter realization inside (3), proves the coordinate discrepancy and pullback together, and exposes the formal proof and acceptance evidence. It can serve as a checked test case for proposed inverse distances and gauge normalizations. The comparison covers the displayed flow and the cited reconstruction and kernel-equivalence results. It is not an exhaustive priority search for this exact bounded-density example or its formalization; no first-result claim is made.

7 Formal statements and reproducibility

The proof is implemented in Lean 4.30.0 with mathlib revision

`c5ea00351c28e24afc9f0f84379aa41082b1188f`.

Cartesian points use `EuclideanSpace` over three real coordinates. The norm is genuinely Euclidean, chronology uses the spatial radius, and all sampling laws are actual measure-theoretic products and pushforwards.

The main export is

`QuantyraNullCone.higher_dimensional_gauge_counterexample3`.

It has no input hypotheses: the explicit parameter is part of the definitions. Its conjunction gives both density-class memberships, all labeled and unlabeled law equalities, a positive infimum over all spatial linear isometries, and the metric pullback for every point and tangent-vector pair. Smooth automorphism and chronology are also exported separately. The following map identifies the key statements, all in namespace `QuantyraNullCone`:

- Volume and probability: `lorentz_diamond_volume3`, `flat_density_class3`, in `LorentzVolume.lean` and `LorentzFlat.lean`.

- Lemma 2.1: `lorentz_flow_smooth_automorphism3`, in `LorentzSmooth.lean`, with inverse and preservation dependencies.
- Lemma 2.2: `hasFDerivAt_lorentz_flow3`, `lorentz_flow_derivative_det3`, `lorentz_flow_derivative_bilinear3` and `lorentz_flow_separation3`, in `LorentzDerivative.lean`, `LorentzJacobian.lean` and `LorentzChronology.lean`.
- Proposition 2.3: `lorentz_flow_chronological_iff3`, in `LorentzChronology.lean`.
- Lemma 3.1 and Proposition 3.2 at a_* : `gauge_density_bounds3`, `gauge_density_lipschitz3`, `gauge_density_class3`, `gauge_forward_transport3`, in the `LorentzGauge` modules and `LorentzTransport.lean`.
- Proposition 4.1: `gauge_order_laws_equal3` and `gauge_unlabeled_order_laws_equal3`, in `LorentzOrderLaws.lean`.
- Proposition 5.1: `gauge_anchor_density_gt3` and `gauge_o2_distance_positive3`, in `LorentzSpatialDistance.lean`.
- Proposition 5.2: `gauge_metric_isometry3`, in `LorentzIsometry.lean`.

The general- a transport statement, displayed sharper density and Lipschitz constants, explicit rational anchor lower bound, Corollary 5.3, and literature parameter conversion are proved in the exposition. The accepted Lean exports give the fixed a_* transport, class bounds, Lipschitz constant two and strict positive distance; we do not claim separate exports for every displayed refinement. Likewise, the interpretation of the coordinate bilinear pullback as a manifold isometry and the determinant calculation of metric volume are spelled out here; the Lean theorem uses the actual Cartesian derivative and bilinear metric rather than a packaged Lorentzian-manifold structure.

The accepted proof/evidence revision is `74c1f743d085c63b45ac2bf30008ad5d29b98fbc`. Authoritative verification ran on GCP under `space-lorentz-acceptance-20261008T071627Z-221cd8`. The root build completed 3001 jobs and 295 selected type/axiom audits across the full repository, including 76 exports added by this campaign. This count includes other research modules. No owned warnings or added axioms were reported. The selected dependencies use only `propext`, `Classical.choice` and `Quot.sound`. Source hashes and all nine pinned dependency identities were checked before and after execution. The raw logs, receipt and failed-development ledger are retained in the repository's `evidence/gcp` directory.

Every development and acceptance Lean invocation for this campaign ran on GCP. Hosted CI is supplementary. This manuscript reuses that acceptance after checking the relevant source identities; it requires no new Lean compilation. Its symbolic and rational algebra checks supplement the proof audit and do not certify analytic or probabilistic claims by testing. The manuscript directory contains build instructions, a formal statement map, primary-source comparison and a review receipt.

8 Implications for a corrected inverse problem

A time-oriented conformal diffeomorphism $H : \mathcal{D} \rightarrow \mathcal{D}$ with $H^*\eta = \Omega_H^2\eta$ acts on volume-normalized density descriptions by

$$\rho_H(p) = |\det dH_p| \sigma(H(p)) = \Omega_H(p)^3 \sigma(H(p)). \quad (16)$$

Whenever the densities in question meet the required regularity, $g_{\rho_H} = H^* g_\sigma$. Omitting the Jacobian factor would transform scalar values, rather than the sampling-volume density. In the example, $H = F_{-a_*}$ and $\sigma = 1$.

An inverse problem for coefficients must therefore justify a normalization that removes this freedom, or use an error quantity on the appropriate geometric equivalence classes. The full conformal action need not preserve the fixed bounds in $\mathcal{K}_3$. Moreover, nonconstant Jacobian weights mean that simply taking an infimum of coefficient differences under that action does not automatically yield a symmetric metric. A corrected definition requires its own invariance and stability analysis.

This example does not supply a finite reconstruction lemma for that corrected problem. The quantitative higher-dimensional inverse question remains open in this project. It also does not alter the separate $1 + 1$ theorem, whose prescribed null-coordinate marginal normalization is a different model assumption. Both metrics here represent the same geometry, so there is no conclusion that two distinct physical spacetimes are indistinguishable by the observations.

The work was developed with OpenAI Codex assistance under the author’s research direction. Quantytra’s review workflow is open-source, decentralized and informal: readers can inspect, use, test and improve the public artifacts. Specialist independent review is not a publication prerequisite; no external review or adoption is claimed by this manuscript.

A Cartesian algebra certificate

Let $c = 1 - a^2$, $D = D_a(p)$, $N = N_a(p)$ and let the numerator vector be $A = (N, cx, cy)$. Its derivatives are

$$\nabla D = (2a(1 + at), -2a^2x, -2a^2y), \quad \nabla N = (1 + a^2 + 2at, -2ax, -2ay).$$

The quotient rule gives

$$(dF_a)_{ij} = \frac{B_{ij}}{D^2}, \quad B = D dA - A(\nabla D)^\top.$$

With $E = \text{diag}(-1, 1, 1)$, expansion gives the polynomial identities

$$B^\top EB = c^2 D^2 E, \quad \det B = c^3 D^3.$$

Division by D^4 and D^6 respectively recovers both identities in (11). These are exactly Cartesian identities, including $x = y = 0$; no limiting polar computation is needed. The supplementary symbolic script checks these polynomials, the inverse and separation numerators, the null-coordinate formulas and the literature parameter conversion. Exact rational checks cover the class constants and the interior anchor. Analytic domain restrictions, smoothness and probability transport are justified in the proofs above and by the mapped Lean statements, rather than by symbolic cancellation.

References

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